Manuscript prepared for Frontiers of Computer Science

RESEARCH ARTICLE

Critical Overflow in Weakly Coupled Two-Choice Allocation

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Abstract

Community totals miss a leading contribution to overflow in two-choice allocation with finite bucket capacity. For two weakly coupled communities of q buckets each, we identify this positive order-q contribution through the internal load distribution. Each arrival joins the less loaded of two candidates. Internal candidates are sampled uniformly across two equal sides, and ties favor the left side. Cross-community sampling has probability ε and uses predetermined kernels with uniform endpoint marginals. Starting from empty buckets, we let $q \to \infty$ with $\varepsilon q \to \lambda < \infty$ and integer capacity c satisfying $2c/q \to \kappa > 0$. We prove a joint limit for community imbalance and the overflow curve on the q^2 arrival scale. A scalar diffusion describes imbalance, while an attracting periodic load profile determines overflow. The diffusion drift averages the common lattice phase. The overflow curve retains the phase at observation. The limit holds for every admissible fixed cross kernel. First Wasserstein convergence also gives convergence of expected overflow divided by q. At nominal total capacity, expected overflow is order q, although its fraction of arrivals tends to zero. The proof combines a coupling of the actual allocation processes with a contraction argument for the internal profile. At zero normalized imbalance, certified evaluation places the limiting overflow coefficient in $[0.63127654, 0.63129462]$. Numerical comparisons separately assess the internal contribution and the diffusion approximation.

Keywords

Randomized allocation; Two choices; Load balancing; Capacity overflow; Diffusion approximation; Universality

1 Introduction

Two-choice allocation assigns each item to the less loaded of two candidates [1]. With finite bucket capacity, the performance question is how many items overflow as the system fills. For two communities of q buckets each, we prove that community imbalance alone misses a positive order-q contribution in the critical limit. The load distribution within each community determines this missing term. We derive a joint critical law that retains both sources of overflow.

The distinction matters even when the two community totals agree. Individual buckets can exceed capacity while others have unused space. At total nominal capacity, overflow equals half the total absolute deviation of bucket loads from capacity. Aggregating loads by community removes part of this deviation. The internal load profile restores the omitted contribution.

Each community uses uniform internal bipartite candidates. A fraction ε of arrivals sample across the communities through any predetermined doubly stochastic kernel. Both fixed matchings and product sampling have the required uniform endpoint marginals. Ties favor the left endpoint, as in the internal two-group Always-Go-Left rule of Vöcking [2].

With empty starts and integer capacity $c \asymp q$, critical fill occurs at $n_* = 2qc \asymp q^2$ arrivals. Without cross sampling, community choices produce order-q imbalance fluctuations. Cross arrivals can contribute order εq to the cumulative normalized drift. These scales motivate the regime $q \to \infty$, $\varepsilon q \to \lambda < \infty$, and $2c/q \to \kappa > 0$.

Our critical law combines a scalar diffusion for community imbalance with an attracting periodic profile for internal loads. At critical fill, let d denote normalized imbalance. The limiting normalized overflow readout is $S_0(d)$, whereas normalized community excess is $|d|/2$. We prove $S_0(d) > |d|/2$ for every finite d. Corollary 2.4 converts this gap into a strictly positive leading correction to expected overflow.

The leading capacity cost is common to every admissible fixed cross kernel. Expected overflow is order q, while its fraction of arrivals tends to zero. Thus, changing a fixed matching to independent product sampling preserves the leading critical overflow coefficient. This conclusion separates the effect of cross-sampling correlations from the internal variation that drives overflow.

Fig. 1 shows the missing internal contribution. Section 6 evaluates the profile readout and the terminal diffusion in separate comparisons.

1.1 Contributions

The main contribution is a joint asymptotic law for capacity overflow under the stated allocation rule and scaling. Theorems 2.1–2.3 establish the following results.

(i) **A joint critical law and its capacity consequences.** Normalized imbalance converges to a scalar diffusion when $\varepsilon q \to \lambda < \infty$. Its restoring drift averages the fast common lattice phase. The overflow curve retains the observation phase and the internal profile. Joint first Wasserstein convergence gives distributional limits and leading mean asymptotics. Corollaries 2.4 and 2.5

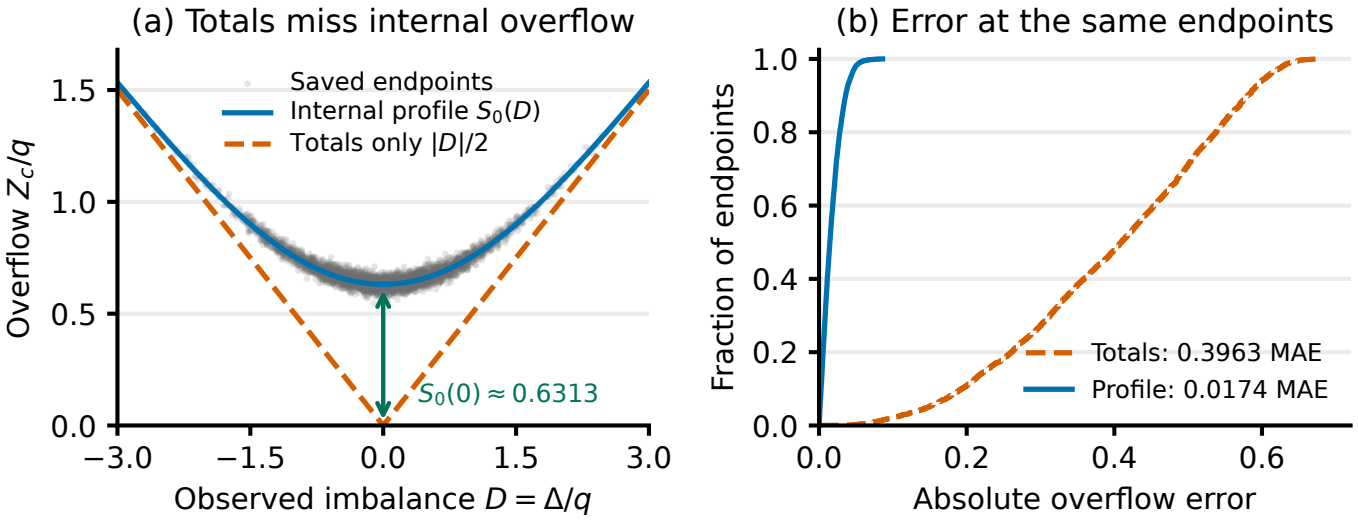


Fig. 1 Community totals miss internal overflow at critical fill. Left: 4096 saved endpoints at $q = 512$, $\kappa = 1$, $\lambda = 1$, with matching crosses. The profile curve retains internal variation; $|D|/2$ uses only community excess. Their gap at $D = 0$ is approximately 0.6313. Right: absolute-error distributions for the same endpoints, with mean absolute errors in the legend. Both readouts use the same observed imbalance, isolating the internal contribution. The frozen profile lookup error is at most 0.000026.

identify the positive internal contribution and the resulting order-q capacity cost.

(ii) **The same leading law for all admissible fixed cross kernels.** For any predetermined doubly stochastic cross kernels, we couple the genuine fixed-kernel process to the genuine product process. On each fixed Tq^2 horizon, their four sorted side vectors have expected uniform ℓ_1 distance $O_{T,\Lambda}(q^{5/6}\log^2 q)$ when $\varepsilon q \le \Lambda$. The same bound controls their differences in imbalance and overflow, simultaneously for all common capacities. After division by q, it vanishes and transfers the critical law to the entire stated kernel class. Random-label forgetting first controls the whole cross load-pair table. A coupling of sorted ranks then preserves the laws of both actual processes.

(iii) **An attracting internal profile with a direct contraction proof.** We construct a unique mean-locked periodic family for the two-side tail flow. It has full support, exponential moments, and uniform attraction on fixed centered exponential-moment sublevels. An exact positive midpoint evolution of signed tail differences gives strict contraction on equal-mean slices. Its half-phase left/right identity records the asymmetric tie rule. This profile supplies both the diffusion drift and the limiting overflow readout.

Certified evaluation places $S_0(0)$ in $[0.63127654, 0.63129462]$. Comparisons using saved allocation outcomes separately assess the internal profile and the terminal diffusion. The profile readout uses no fitted parameter.

1.2 Relation to previous work

Existing rules and distribution equations. Vöcking [2] introduced Always-Go-Left. Berenbrink, Czumaj, Steger, and Vöcking [3] give sharp heavy-load deviation bounds for greedy and asymmetric allocation. Mitzenmacher [4] uses differential equations to describe full load distributions. We use this established distribution-level viewpoint. The additional requirements here are long-time attraction of a two-type integer-lattice profile and its coupling to a slow community imbalance.

Correlated choices and process comparison. Mitzenmacher [5, 6] shows asymptotic equivalence between double hashing and fully random balanced allocation, using fluid limits and a coupling through a biased intermediate process. The stated distribution results concern a constant number of choices and TN arrivals into N bins for fixed T. Our comparison treats arbitrary fixed doubly stochastic cross kernels, allows only order-q^{-1} cross probability, and covers Tq^2 arrivals. It bounds the distance between the genuine fixed-kernel and product processes. These differences are needed before passing to the critical overflow law.

Graphs and queues. Peres, Talwar, and Wieder [7] study graphical allocation and limited choice. Bansal and Feldheim [8] give a strategy based on graph decompositions. Our rule compares only the two endpoint loads. Queueing universality and diffusion limits [9–11] include service and use different scales. Here each item has two candidates and loads only increase.

Traveling profiles. Stolyar [12, 13] develops traveling-wave and stationary-limit methods. Contraction and compactness are precedents. The existence result in [12] permits general jumps. Its uniqueness and attraction result requires a positive jump density bounded away from zero on compact subsets of $\mathbb{R}_+$. The arithmetic-jump caveat following that theorem prevents its direct use for the present lattice dynamics. We prove the two-side periodic-family result directly, then obtain finite-system tracking without a stationary-limit interchange. The later phase averaging acts on the slow drift, while the critical overflow retains its endpoint phase.

Sections 2–5 state and prove the results. Section 6 evaluates their prediction steps. The supplement gives numerical certificates, statistical protocols, and provenance.

2 Model and main results

There are communities $C_0 = L_0 \cup R_0$ and $C_1 = L_1 \cup R_1$, each side of size m, and $q = 2m$. At each arrival choose an internal edge in C_s with probability $(1-\varepsilon)/2$, for each s, or direction L_0–R_1 or L_1–R_0 with probability $\varepsilon/2$ each. Internal edges are uniform on $K_{m,m}$. In cross direction d, a fixed nonnegative matrix $B^{(d)}$ has all row and column sums one; conditional edge probability is $B^{(d)}_{ij}/m$. Product sampling has $B = J/m$ and a matching is a permutation matrix. Matrices may depend deterministically on q. A randomly selected matrix is admissible if chosen independently before the arrivals, by conditioning on it. Matrices selected from current loads are excluded.

Conditional on the fixed matrices, event types and candidate draws are independent across arrivals. Loads $A_v(n)$ are nonnegative integers, initially zero. Increment the endpoint of smaller load, breaking every tie to the left side. Let

$$M_s(n) = \sum_{v \in C_s} A_v(n), \quad \Delta_n = M_0(n) - M_1(n),$$
$$Z_c(n) = \sum_v (A_v(n) - c)_+.$$

For integer c, Z_c is also the number of overflows when each bucket stores its first c assigned items. Indeed the capped loads remain $\min(A_v, c)$ under the same candidate stream: below capacity both comparisons agree, and once both candidates are full the uncapped winner is irrelevant to the overflow count. This identifies the overflow count with the positive-part load functional.

For a pair of integer laws $z = (p_L, p_R)$ with finite first moments, write

$$x_k = p_L([k,\infty)), \quad y_k = p_R([k,\infty)),$$
$$a_k = x_k - x_{k+1}, \quad b_k = y_k - y_{k+1},$$

and $\mu(z) = (\mathbb{E}L + \mathbb{E}R)/2$. The metric d_1 is the sum of the two marginal Wasserstein distances, equivalently the sum of absolute tail differences. Integer shift S_j adds j to both laws.

There are three time scales. After k internal arrivals in one community, its mean clock is $u = k/q$. The common mean at global arrival n is $h_n = n/(2q)$. The slow diffusion clock is $t = n/q^2$. At critical fill $n_* = 2qc = 4mc$, these give $h_{n_*} = c$ and $n_*/q^2 = c/m \to \kappa$. An offset of $2hq$ arrivals changes the common mean by h while changing slow time by only $2h/q$. Fig. 2 shows the candidate structure and the two retained states.

For integer capacities c_q, set

$$n_* = 2qc_q, \qquad n_q(h) = n_* + \lfloor 2qh \rfloor.$$

When $c_q/m \to \kappa > 0$, each bounded h window has nonnegative indices for large q. The imbalance path D_q linearly interpolates Δ_n/q at times n/q^2. The overflow curve F_q linearly interpolates $Z_{c_q}(k)/q$ at coordinates $h = (k - n_*)/(2q)$. Its distance from $Z_{c_q}(n_q(h))/q$ is at most $1/q$.

All path and curve Wasserstein distances below have order one. They use the uniform norm on the corresponding continuous-function space. Joint convergence uses the sum of the path and curve uniform norms. Thus, for a metric d, the transport distance is $\inf \mathbb{E}d(U,V)$ over couplings with the stated laws.

Theorem 2.1 (Fixed-kernel universality)**.** For empty starts and fixed $T, \Lambda < \infty$, assume $q \geq 4$, $\varepsilon \leq 1/4$ and $\varepsilon q \leq \Lambda$. A genuine fixed-kernel process and a genuine product process admit a coupling such that, for $\mathcal{D}_n$ the sum of absolute differences of their four separately sorted side vectors,

$$\mathbb{E} \max_{n \leq Tq^2} \frac{\mathcal{D}_n}{q} = O_{T,\Lambda}(q^{-1/6} \log^2 q) \longrightarrow 0.$$

The same bound controls $|\Delta^F - \Delta^P|/q$ and $|Z^F_c - Z^P_c|/q$, simultaneously for all common capacities $c \geq 0$.

Theorem 2.2 (Internal periodic profile)**.** The mean-clock tail flow

$$\dot{x}_k = 2a_{k-1}y_{k-1}, \qquad \dot{y}_k = 2b_{k-1}x_k \tag{2.1}$$

has a unique finite-first-moment periodic family $\rho^u = (L_u, R_u)$ with mean u, satisfying

$$\rho^{u+1} = \mathsf{S}_1 \rho^u, \qquad \Phi_t \rho^u = \rho^{u+t}.$$

It has exponential moments, full support at every integer, and $d_1(\rho^u, \rho^v) = 2|u-v|$. Attraction is uniform over centered exponential-moment sublevels and all initial phases.

Define independent-draw comparisons and the overflow readout by

$$\begin{aligned} H(u,v) &= \mathbb{P}(L_u \leq R_v), \\ g(h,d) &= H(h + d/2, h - d/2) - H(h - d/2, h + d/2), \\ \overline{g}(d) &= \int_0^1 g(h,d)\,\mathrm{d}h, \\ \sigma(u) &= \tfrac{1}{2}\{\mathbb{E}[(L_u)_+] + \mathbb{E}[(R_u)_+]\}, \\ S_h(d) &= \sigma(h + d/2) + \sigma(h - d/2). \end{aligned} \tag{2.2, 2.3}$$

Theorem 2.3 (Common critical limit)**.** If $\varepsilon q \to \lambda \in [0,\infty)$, then for every fixed T, $D_q \to D$ in first Wasserstein distance for the uniform path norm on $[0,T]$, where

$$\mathrm{d}D_t = \mathrm{d}B_t + \lambda \overline{g}(D_t)\,\mathrm{d}t, \qquad D_0 = 0. \tag{2.4}$$

The function $\overline{g}$ is odd, smooth, strictly decreasing, globally 2-Lipschitz and negative on $(0,\infty)$, with $\overline{g}'(0) < 0$ and $\overline{g}(d) \to -\operatorname{sign}(d)$ at infinity. Let the integer capacities satisfy $c_q/m \to \kappa \in (0,\infty)$. For each $H_0 < \infty$ and $T_* > \kappa$,

$$\big(D_q|_{[0,T_*]}, F_q|_{[-H_0,H_0]}\big) \longrightarrow \big(D|_{[0,T_*]}, (S_h(D_\kappa))_{|h| \leq H_0}\big)$$

in first Wasserstein distance for the sum of the two uniform norms. Restriction gives joint convergence with any shorter path horizon. This holds for every fixed kernel above. The function S_h is even and convex in d, 1-Lipschitz in d, and 2-Lipschitz in h, with $S_0(0) > 0$.

The drift averages the common phase along many complete cycles. The overflow readout at the integer critical count retains phase zero. The observation offset h changes both the fill level and the lattice phase, so S_h retains both effects.

Fig. 3 displays the profile and averaged response.

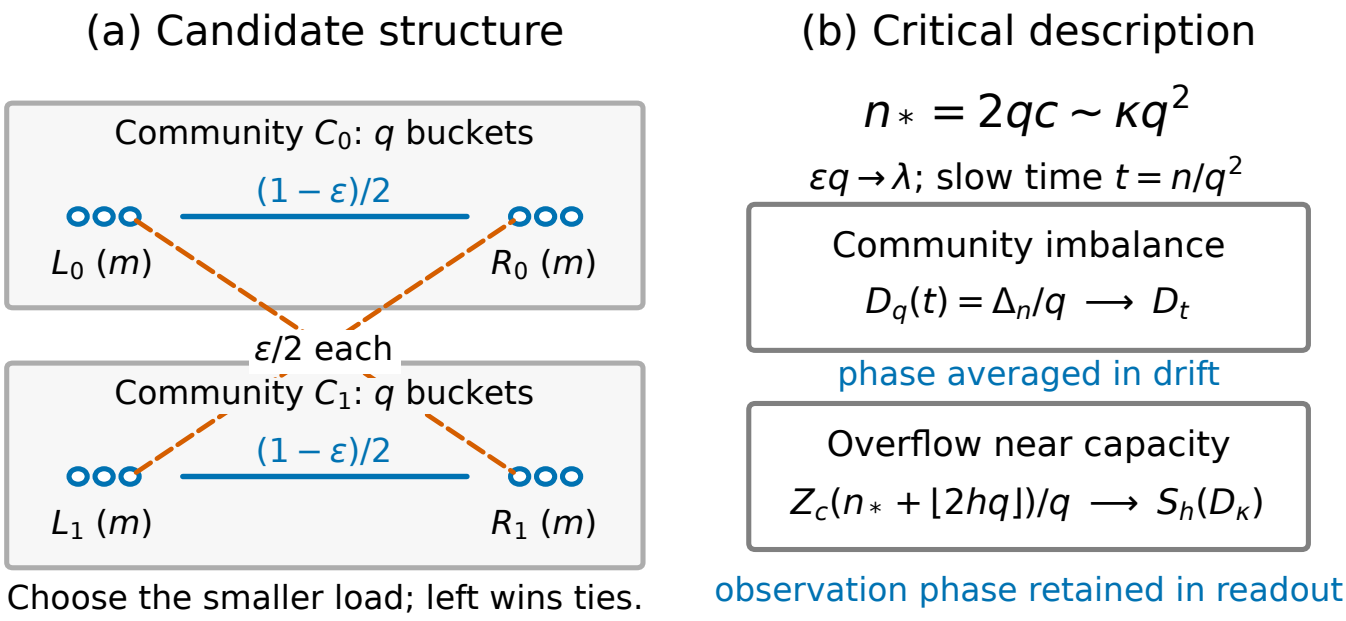


Fig. 2 The fixed allocation rule and its critical scales. Each of the four sides contains $m = q/2$ buckets. Internal directions have probability $(1-\varepsilon)/2$ each; cross directions have probability $\varepsilon/2$ each. The offered cross endpoints have uniform marginals under any fixed doubly stochastic kernel. The algorithm increments the smaller endpoint load and chooses the left endpoint at a tie. The limiting diffusion retains community imbalance; the overflow readout also retains the internal profile.

Corollary 2.4 (A leading internal overflow contribution)**.** Under Theorem 2.3, fix $\kappa > 0$ and $\lambda < \infty$. Let X_u have the equal mixture of the two laws in ρ^u, and set

$$r(u) = \mathbb{E}[(X_u)_+] - u_+ = \tfrac{1}{2}(\mathbb{E}|X_u| - |u|).$$

Then $r(u) > 0$ for every finite u, and

$$S_0(d) = \frac{|d|}{2} + r(d/2) + r(-d/2).$$

For every admissible fixed cross kernel,

$$\lim_{q\to\infty} \mathbb{E}\left[\frac{Z_{c_q}(n_*) - |\Delta_{n_*}|/2}{q}\right] = \mathbb{E}\{r(D_\kappa/2) + r(-D_\kappa/2)\} > 0.$$

The positive limit may depend on κ and λ.

Proof. The mixture has mean u. Full support from Theorem 2.2 gives positive mass on both sides of zero. Hence $\mathbb{E}|X_u| > |\mathbb{E}X_u| = |u|$. Substitution into (2.3) gives the displayed decomposition.

At $n_* = 2qc_q$, the community excess totals are $\Delta_{n_*}/2$ and $-\Delta_{n_*}/2$. Taking positive parts within each community yields the exact finite-size inequality $Z_{c_q}(n_*) \geq |\Delta_{n_*}|/2$. Put $\kappa_q = n_*/q^2 \to \kappa$. The endpoint-time estimate (5.11) shows $\mathbb{E}|D_q(\kappa_q) - D_q(\kappa)| \to 0$. The joint first Wasserstein convergence in Theorem 2.3 therefore passes expectations for the Lipschitz, linear-growth difference readout. This gives the stated limit. Its integrand is strictly positive at every finite value of D_κ, and its expectation is finite. □

The two r terms identify the internal contribution omitted by a community-excess readout. The profile preserves this contribution at each fixed $\kappa > 0$ and $\lambda < \infty$.

Corollary 2.5 (Capacity consequences)**.** Under Theorem 2.3, let $s_* = S_0(0) > 0$ and $C_{\kappa,\lambda} = \mathbb{E}S_0(D_\kappa)$. For each fixed $a \in (0, s_*)$,

$$\mathbb{P}\{Z_{c_q}(n_*) \geq aq\} \longrightarrow 1, \qquad \mathbb{P}\{Z_{c_q}(n_*) = 0\} \longrightarrow 0.$$

Moreover,

$$\mathbb{E}Z_{c_q}(n_*) = qC_{\kappa,\lambda} + o(q), \qquad \frac{\mathbb{E}Z_{c_q}(n_*)}{n_*} = \frac{C_{\kappa,\lambda}}{\kappa q} + o(q^{-1}).$$

These conclusions and their leading coefficient hold for every admissible fixed cross kernel.

Proof. Evenness and convexity give $S_0(d) \geq S_0(0) > a$ for every d. Weak convergence of $Z_{c_q}(n_*)/q$ and the open-set inequality therefore give the probability limits. First Wasserstein convergence passes its expectation to $C_{\kappa,\lambda} < \infty$. Finally, $n_*/q^2 \to \kappa > 0$ gives the fractional asymptotic. The common limit is independent of the fixed cross kernel. □

The leading overflow cost is common to all fixed cross kernels in the stated class.

■ 3 From a fixed cross kernel to product sampling

This section proves Theorem 2.1 by coupling the actual processes. Internal sampling first forgets a random label transposition under a dispersion bound. This controls the whole cross load-pair table. A sorted-rank coupling then uses greedy nonexpansion. Fix $\theta = 1/200$, $K_0 = 16/3$, and $I = (1-\varepsilon)/2$.

Lemma 3.1 (Dispersion)**.** With $\xi_v = A_v - M_{s(v)}/q$ and $\Gamma = \sum_v \cosh(\theta\xi_v)$, every fixed uniform-marginal kernel satisfies

$$\sup_{n\geq 0} \mathbb{E}\Gamma_n \leq K_0 q. \tag{3.1}$$

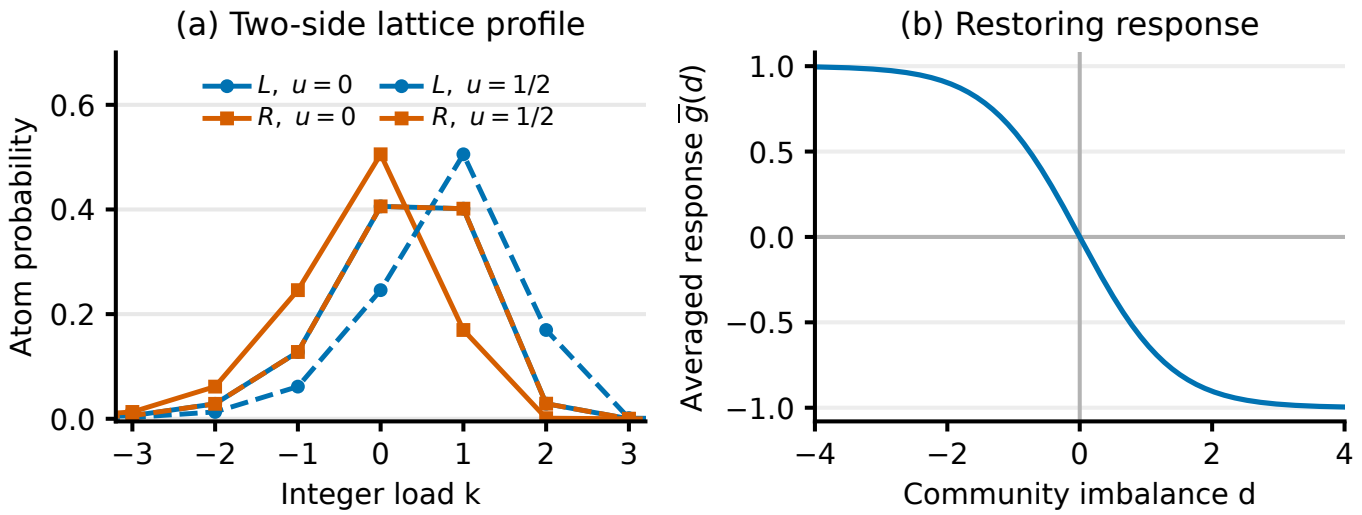


Fig. 3 Internal structure and restoring response, replotted from saved profile nodes and the frozen response table. Left: left and right atom probabilities at mean phases $u = 0$ (solid) and $u = 1/2$ (dashed). Only the displayed load levels are shown. The half-phase operation swaps the sides and shifts the new left law by one. Right: the phase-averaged response $\overline{g}(d)$; positive imbalance has negative response. Connecting segments join the saved nodes.

Proof. The rank-based potential argument is given in Supplement section SM2.1. It uses only the cross marginals. □

Lemma 3.2 (Random transposition)**.** Let μ_n be the law of the actual fixed-kernel configuration. Independently choose two uniform labels i, j in one side, and let τ swap them. For $\ell \geq 4$, integer $L = q\ell$, and $n \geq L$,

$$\mathbb{E}_{i,j}\,\mathrm{TV}(\mu_n, \tau\mu_n) \leq \eta, \tag{3.2}$$

$$\eta = \min\left\{1, \frac{32K_0}{\theta\sqrt{\ell}} + 16K_0 e^{-\theta\ell/16} + e^{-L/48} + 4\varepsilon\ell\right\}.$$

Proof. Compare the original and transposed configurations through the conjugate kernel. The two discrepancies behave as tags until they meet. Suppress cross events touching a tag; internal updates then make the tag gap no harder to absorb than a fair walk. Dispersion controls the initial gap and deviations at the window endpoints. On the remaining event, nonmeeting requires at least $\lceil \ell/4 \rceil$ embedded jumps. A fixed-jump reflection bound gives the first term of η. The conjugate-kernel comparison adds a second cross-touch allowance. Supplement section SM2.4 gives the full coupling and constants; it does not condition a martingale estimate on a future jump count. □

For one cross direction write Q_n for its actual conditional joint load-pair law and p_n, r_n for the empirical side marginals. If $0 \leq f \leq 1$, define

$$C_B(f) = \frac{1}{m}\sum_{ij} B_{ij} f(U_i, V_j) - \frac{1}{m^2}\sum_{ij} f(U_i, V_j).$$

The test may use parameters unchanged by the side swap, including community means. Let I_0 be uniform, draw $J_0 \mid I_0$ from the corresponding row of B, and draw J_0' independently and uniformly. We claim

$$\mathbb{E}C_B(f)^2 \leq 2\eta + 2/m, \qquad \mathbb{E}|C_B(f)| \leq \sqrt{2\eta + 2/m}. \tag{3.3}$$

Write $C = C_B(f)$ and take these auxiliary indices independently of the configuration. Pointwise,

$$C^2 = \mathbb{E}_{\mathrm{aux}}\left[C\{f(U_{I_0}, V_{J_0}) - f(U_{I_0}, V_{J_0'})\}\right].$$

For fixed indices let τ exchange J_0, J_0'. The product-marginal term of C is unchanged, and $|C(\tau A) - C(A)| \leq 2/m$. Apply total variation to $G(A) = C(A) f(U_{I_0}, V_{J_0})$, for which $|G| \leq 1$. Thus the expected difference is at most $2\,\mathrm{TV}(\mu_n, \tau\mu_n) + 2/m$. After averaging, (J_0, J_0') are independent uniform side labels: column sums make J_0 uniform, even though it depends on I_0. Lemma 3.2 gives (3.3). The bounded-test TV cost is 2η. This controls the second moment, rather than only a signed average.

Center each load coordinate at its community's integer mean part. For each fixed offset cell in $[-R, R]^2$, its indicator satisfies (3.3). Summing over this deterministic square and using (3.1) outside it yields

$$\mathbb{E}\,\|Q_n - p_n \otimes r_n\|_1 \leq \rho,$$
$$\rho = \min\{2, (2R+1)^2\sqrt{2\eta + 2/m} + 16K_0 e^{-\theta(R-1)}\}. \tag{3.4}$$

Indeed Q and the product have the same marginals; twice their marginal tail masses bounds the mass outside the square. Replacing the mean by its integer part costs at most one in the exponential bound. No supremum over arbitrary data-dependent tests is taken.

It remains to compare two actual process laws. Sort each side separately. Integer greedy updates using the same offered ranks are nonexpansive in sorted ℓ_1 distance: if different winners are chosen, the two relevant coordinate differences have the ordering needed for their $+1, -1$ changes not to increase the absolute sum; sorting again minimizes the matching cost. Internally use common independent uniform ranks. At a cross, sample the actual fixed edge in the fixed process, then maximally couple its load pair from Q_F to a virtual pair from $p_F \otimes r_F$. Given the virtual loads, choose ranks independently and uniformly inside the equal-load blocks. Conditional on the entire joint past before

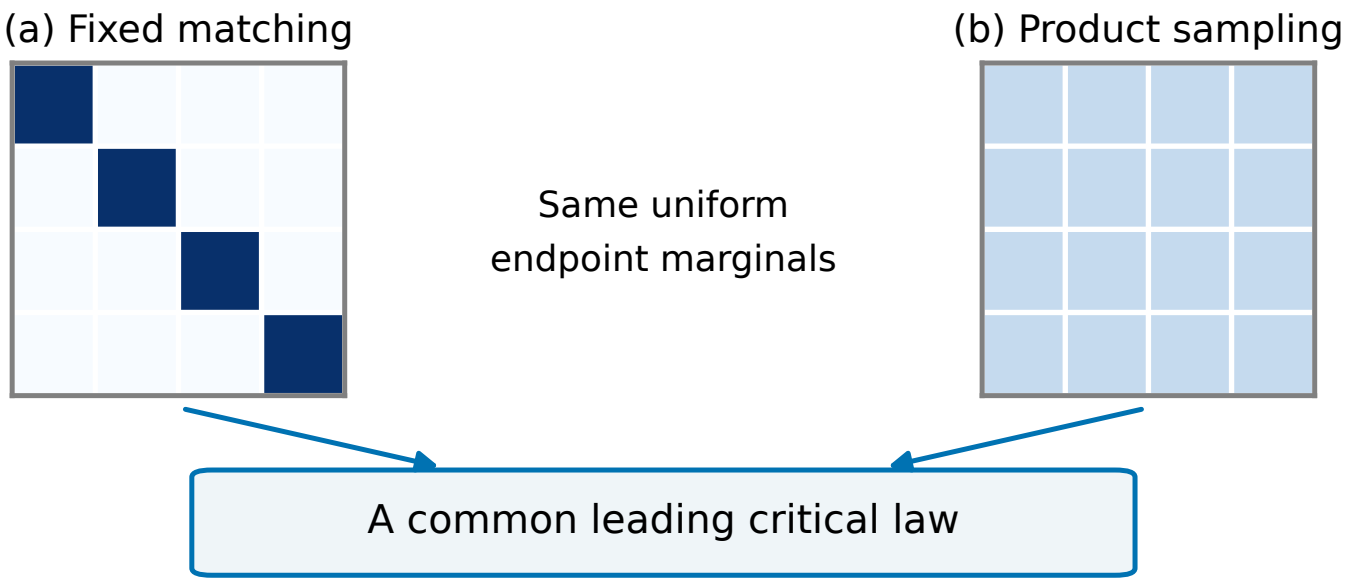


Fig. 4 Different fixed cross correlations share the critical law under the model and scaling of Theorems 2.1–2.3. The grids show conditional candidate-pair probabilities in one cross direction, with four illustrative labels on each side. Matching probabilities are $1/4$ on the diagonal and zero elsewhere; product probabilities are all $1/16$. Both have uniform marginals. Internal sampling and rare cross arrivals support the process comparison.

drawing the current edge, these ranks are independent uniforms, since a block of size k has mass k/m followed by probability $1/k$ per rank. Use the same ranks in the product process, which therefore has its genuine transition law. Independence need not persist after additionally conditioning on the current realized fixed edge.

On successful load-pair coupling the actual and virtual sorted fixed-process updates agree, including ties. On failure their distance is at most two. Hence

$$\mathcal{D}_{n+1} \le \mathcal{D}_n + 2\mathbf{1}_{\{\text{cross and load-pair coupling fails}\}}.$$

Before L use failure probability one; afterwards its expected conditional value is at most $\rho/2$. Summing the nonnegative counts gives the explicit bound

$$\mathbb{E}\max_{n\le N}\mathcal{D}_n \le 2\varepsilon\min(N,L) + \varepsilon\rho(N-L)_+. \tag{3.5}$$

Take $\ell = \max(4, \lceil q^{2/3}\rceil)$ and $R = \lceil (2/\theta)\log q\rceil + 1$. Then (3.5) divided by q, for $N \le Tq^2$, is $O(q^{-1/6}\log^2 q)$. Both the community difference and every positive-part sum are 1-Lipschitz in sorted load distance. This proves Theorem 2.1.

4 The mean-locked internal orbit

The main step is an exact midpoint equation for the difference of two tail flows. It gives strict contraction on equal-mean slices without uniform irreducibility on the whole lattice. Compactness then yields the periodic family and uniform attraction on each fixed moment sublevel. In this section all times use the mean clock. Signed lattice coordinates express translated laws; actual allocations remain nonnegative. Equation (2.1) has a unique probability flow on finite-first-moment laws. It can be constructed by two coupled bounded-rate birth laws with left rate $2y_k$ and right rate $2x_{k+1}$. Picard iteration, or finite truncation followed by tightness, gives existence and uniqueness. The tail field is globally Lipschitz, probability constraints are invariant, and rates at most two prevent explosion. The sum of the mean derivatives is two, so $\mu(\Phi_t z) = \mu(z) + t$.

4.1 Moment compactness and exact nonexpansion

Let

$$V_\theta(z) = \mathbb{E}\cosh(\theta(L-\mu)) + \mathbb{E}\cosh(\theta(R-\mu)).$$

The selected-load distribution is $w_k = a_k y_k + b_k x_{k+1}$, of total mass one, bounded by twice the pooled law $\nu = (p_L + p_R)/2$. Its suffix is $x_k y_k \le [(x_k + y_k)/2]^2$. The same rank/Gini argument as in Lemma 3.1, now with pure-internal rank deficit one, and differentiation of the moving mean give

$$\frac{\mathrm{d}}{\mathrm{d}t}V_\theta(\Phi_t z) \le -\{\theta/2 - 2(e^\theta - 1 - \theta)\}V_\theta + \theta \le -\theta V_\theta/4 + \theta. \tag{4.1}$$

Truncation justifies this calculation for finite exponential moments. Thus every centered sublevel $V_\theta \le M$, $M \ge 4$, is invariant. This moment bound supplies the invariant sublevels used in the attraction argument.

For two solutions put $d_k = x_k - \widetilde{x}_k$, $e_k = y_k - \widetilde{y}_k$, and let bars denote midpoint tails or atoms. Their difference satisfies exactly the forward equation on states X_k, Y_k with transitions

$$\begin{aligned} X_k &\longrightarrow X_{k+1} \quad \text{at rate } 2\bar{y}_k, & X_k &\longrightarrow Y_k \quad \text{at rate } 2\bar{b}_{k-1},\\ Y_k &\longrightarrow Y_{k+1} \quad \text{at rate } 2\bar{x}_{k+1}, & Y_k &\longrightarrow X_{k+1} \quad \text{at rate } 2\bar{a}_k. \end{aligned}$$

Exit rates are $2\bar{y}_{k-1}$ and $2\bar{x}_k$, at most two. The midpoint identity is exact for the bilinear field. Its evolution is positive and mass preserving, so

$$d_1(\Phi_t z, \Phi_t \widetilde{z}) \le d_1(z, \widetilde{z}). \tag{4.2}$$

It also shows that the field is 4-Lipschitz in tail ℓ_1.

4.2 Strictness on an equal-mean slice

Supplement section SM8.2 derives explicit lower bounds $Ct^p e^{-\gamma t}$ for all sufficiently high atoms, starting from one positive atom on each side. Consequently all finite tails are positive for $t > 0$. On a compact subinterval of $(0, t)$, any two finite midpoint states can follow finite paths to a common high state X_K. The ordered-jump bound in Supplement section SM8.3 makes both transition probabilities $P_{0,t}(i, X_K), P_{0,t}(j, X_K)$ positive. These bounds depend on the flows and chosen states.

For equal means the initial signed tail difference $v = (d, e)$ has total mass zero. If $v \neq 0$, choose $v_i > 0$ and $v_j < 0$. Positivity and mass preservation of the transition kernel give the explicit cancellation bound

$$\begin{aligned}\|v(t)\|_1 &\le \|v(0)\|_1 \\ &\quad - 2\min\{v_i P_{0,t}(i, X_K), (-v_j) P_{0,t}(j, X_K)\} \\ &< \|v(0)\|_1.\end{aligned}$$

Thus for every $t > 0$,

$$d_1(\Phi_t z, \Phi_t \widetilde{z}) < d_1(z, \widetilde{z}) \quad \text{if } \mu(z) = \mu(\widetilde{z}),\ z \neq \widetilde{z}. \tag{4.3}$$

4.3 Existence, uniqueness and uniform attraction

Define $T = \mathsf{S}_{-1}\Phi_1$ and $C_{\alpha,M} = \{\mu = \alpha, V_\theta \le M\}$, $M \ge 4$. Exponential tails and lower semicontinuity make this set compact in d_1. Identical mixtures on $\{0, 1\}$ show nonemptiness for $\alpha \in [0, 1]$. The moment estimate shows that T preserves each slice. Supplement sections SM8.1 and SM8.4 give the full details of these steps and the following fixed-point argument.

Along an orbit, $d_1(T^n z, T^{n+1} z) \downarrow r$. A convergent subsequence with limit w gives $d_1(w, Tw) = d_1(Tw, T^2 w) = r$. Strictness forces $Tw = w$ and $r = 0$. Two fixed points contradict strictness. For any orbit, a subsequential limit of its decreasing distance to w would give the same equality unless that distance tends to zero. Thus every orbit converges to the unique fixed point on its slice.

Let z_* be the fixed point in $C_{0,4}$ and define

$$\rho^u = \mathsf{S}_{\lfloor u \rfloor} \Phi_{\{u\}} z_*.$$

The fixed-point identity gives the flow and period identities, continuity across integer phases, and $V_\theta(\rho^u) \le 4$. Strictness excludes any other finite-first-moment periodic family, including families without an assumed exponential moment.

For fixed $M \ge 4$, the set $K_M = \{z : \mu(z) \in [0, 1], V_\theta(z) \le M\}$ is compact. The functions $f_n(z) = d_1(T^n z, \rho^{\mu(z)})$ are continuous, decrease pointwise to zero, and satisfy $f_{n+1} \le f_n$ by nonexpansion. For $\delta > 0$, the increasing open sets $\{f_n < \delta\}$ cover K_M. A finite subcover has a largest index N; hence $\sup_{K_M} f_n \le \delta$ for $n \ge N$. Nonexpansion for the fractional part of time then gives

$$\omega_M(t) := \sup_{\substack{\mu(z) \in [0,1] \\ V_\theta(z) \le M}} d_1(\Phi_t z, \rho^{\mu(z)+t}) \longrightarrow 0. \tag{4.4}$$

This uses compactness for each fixed M; no uniform minorization is required. Integer translation covers arbitrary means. Along the birth flow each side increases stochastically. Summing the two mean increments therefore gives

$$d_1(\rho^u, \rho^v) = 2|u - v|. \tag{4.5}$$

For any integer k, positivity at $k + n$ after a sufficiently large integer time n transfers to the atom at k by the period relation. This gives full support and completes Theorem 2.2. Further geometric identities are developed next.

4.4 Interlacing, a single front, and the response

The empty-start flow has invariant inequalities $x_k \ge y_k \ge x_{k+1}$. At a boundary equality its gap derivative is nonnegative, directly from (2.1); truncation extends the check to the full lattice. Attraction of centered integer-time empty profiles transfers the inequalities to the orbit. The map $J(p_L, p_R) = (\mathsf{S}_1 p_R, p_L)$ commutes with the flow, has $J^2 = \mathsf{S}_1$, and raises the mean by $1/2$. Uniqueness implies

$$\rho_L^{u+1/2} = \mathsf{S}_1 \rho_R^u, \qquad \rho_R^{u+1/2} = \rho_L^u.$$

Set $z_{2k} = x_k$, $z_{2k+1} = y_k$. The equations and half-phase identity yield a single normalized front F:

$$\begin{gathered}x_k(u) = F(2k - 2u), \quad y_k(u) = F(2k + 1 - 2u), \\ -F'(r) = F(r - 1)[F(r - 2) - F(r)].\end{gathered} \tag{4.6}$$

It decreases strictly from one to zero, is smooth by repeated differentiation, and its translation is fixed by the mean-zero normalization.

The function $w(r) = [F(r) - F(r + 2)]/2$ is a positive density of mass one. Expanding H as its atom–tail sum and substituting $r = 2k - 2h - d$ in each term gives

$$\overline{g}(d) = \int_{\mathbb{R}} w(r)[F(r + 1 + 2d) - F(r + 1 - 2d)]\,\mathrm{d}r. \tag{4.7}$$

The phase intervals tile the real line; Tonelli applies to the original nonnegative sums. Equation (4.7) proves oddness, strict decrease and the far-field limit by dominated convergence. Bounded derivatives from (4.6) give smoothness. Marginal optimal couplings and (4.5) give the useful global estimate

$$|g(h, d) - g(h', d')| \le 4|h - h'| + 2|d - d'|. \tag{4.8}$$

An integer comparison changes by at most the integer displacement.

Differentiating the absolutely convergent atom sums gives

$$\partial_1 H(u, u) = -2\sum_k a_k y_k b_k, \qquad \partial_2 H(u, u) = 2\sum_k a_k x_k b_{k-1}.$$

Integration of the derivative of periodic $H(u, u)$ equates the two sums after phase integration. Thus the restoring coefficient is

$$\alpha = -\overline{g}'(0) = 4\int_0^1 \sum_k a_k(u) y_k(u) b_k(u)\,\mathrm{d}u > 0. \tag{4.9}$$

This also equals $-4\int w(r)F'(r + 1)\,\mathrm{d}r$. Finally the orbit moment bound implies, by a union bound on centered deviations,

$$|\overline{g}(d) + \operatorname{sign}(d)| \le \min(2, 32e^{-\theta|d|/2}), \qquad d \neq 0.$$

This describes the far-field behavior of the fixed drift.

5 Tracking, deterministic averaging and critical overflow

A fixed fluid-time limit does not cover q^2 arrivals, so we restart on fixed windows to obtain tracking at deterministic times. Blocks of $2q$ arrivals then average the common phase in the drift. The endpoint overflow keeps its observation phase; a finite mesh gives uniform convergence of its curve. Work with product cross sampling until the final transfer. Discarding finitely many q, choose $\Lambda < \infty$ such that $\varepsilon q \le \Lambda$ and $\varepsilon \le 1/4$.

5.1 A uniform estimate on a fixed fluid window

Consider one pure community, with $q = 2m$ buckets. Let X_k^q be its pair of empirical side laws after k internal arrivals. The mean clock is $u = k/q$. Interpolate linearly between these times. Start the fluid $X(u) = \Phi_u X_0^q$ from the exact empirical laws. Assume

$$V_\theta(X_0^q) \le R, \qquad \theta = 1/200.$$

Translate all loads by $-\lfloor \mu(X_0^q) \rfloor$. This leaves the metric and dynamics unchanged. The translated initial mean belongs to $[0, 1)$. Fix an integer $B \ge 1$ and a cutoff $K_1 \ge 1$. Then

$$\mathbb{E} \sup_{0 \le u \le B} d_1(X^q(u), X(u)) \le \eta_q(B, R, K_1). \tag{5.1}$$

The bound is uniform over all empirical initial states satisfying the displayed moment condition.

Supplement section SM2.2 gives an explicit η_q and the estimates proving (5.1). With

$$K_1(q) = \lceil 2 \log(q)/\theta \rceil,$$

the bound tends to zero for each fixed B, R. No growing fluid horizon is used.

5.2 Moments, cross suppression, and deterministic restarts

For the real product process, let W_n be the sum of four side distances to orbit laws at the actual community means. Let Ψ_n sum the four empirical cosh moments centered at these means. Lemma 3.1 and Supplement section SM2.3 give

$$\sup_n \mathbb{E}\Psi_n \le C_E = 32/3, \qquad \sup_n \mathbb{E}W_n^2 \le C_*^2 = (32C_E + 128)/\theta^2. \tag{5.2}$$

Start an auxiliary chain from the real state at a deterministic time. Use the same internal candidates, but suppress every cross update. Greedy integer updates with the same candidates are nonexpansive in labelled ℓ_1. If $N_\times$ crosses occur during a window, the total labelled discrepancy is at most $N_\times$. Normalizing each side by $m = q/2$ gives profile distance at most $2N_\times/q$. Each suppressed cross removes one unit of community mass. The sum of the two mean discrepancies is therefore $N_\times/q$. Equation (4.5) adds at most another $2N_\times/q$ for the two orbit targets. Consequently

$$W_n \le W_n^{\mathrm{aux}} + 4N_\times/q. \tag{5.3}$$

Fix $\delta > 0$ and $R \ge 4$. Choose a fixed integer $A \ge 1$ such that $\omega_R(u) \le \delta/2$ for all $u \ge A$. Put

$$L = 2q(A + 1), \qquad B = 2(A + 1) + 1.$$

For any deterministic endpoint $n \ge L$, restart at $n - L$. A start is good if $\Psi_{n-L} \le R$. Its complement has probability at most C_E/R.

Let N_s be the internal event count for community s in the window. Its marginal law is $\mathrm{Bin}(L, (1-\varepsilon)/2)$. For sufficiently large q, its mean exceeds $Aq + q/2$, while its variance is at most $L/4$. Chebyshev and a union bound give

$$\mathbb{P}\{\min(N_0, N_1) < Aq\} \le C_A/q, \qquad C_A = 4(A + 1).$$

Also $N_s \le L < Bq$. Independence between N_0 and N_1 is not needed. The internal candidate streams can be chosen independently of all event-type clocks and the restart state. On good starts, the supremum in (5.1) therefore covers every possible terminal clock. This avoids conditioning a martingale estimate on a future count.

On good starts and clocks, the auxiliary fluid targets are within total distance δ of the orbit. The two empirical approximation errors contribute at most $2\eta_q(B, R, K_1)$. On bad starts or clocks, use Cauchy–Schwarz for the real W_n, with (5.2). Taking expectations in (5.3) yields, uniformly over deterministic $n \ge L$,

$$\mathbb{E}W_n \le \delta + 2\eta_q(B, R, K_1) + 4\varepsilon L/q + C_*\sqrt{C_E/R} + C_*\sqrt{C_A/q}. \tag{5.4}$$

First fix δ, R, then choose A, B. Next let $q \to \infty$. Finally let $R \to \infty$ and $\delta \to 0$. This proves

$$\lim_{H\to\infty} \limsup_{q\to\infty} \sup_{n \ge Hq} \mathbb{E}W_n = 0, \qquad q^{-2} \sum_{n < Tq^2} \mathbb{E}W_n \longrightarrow 0. \tag{5.5}$$

For the second conclusion, the first Hq terms contribute at most C_*H/q. The remaining terms are bounded by $T \sup_{n \ge Hq} \mathbb{E}W_n$. In particular, $\mathbb{E}W_{n_q} \to 0$ whenever the deterministic times satisfy $n_q/q^2 \to t > 0$. The conclusion controls expected profile error at deterministic times.

5.3 Exact imbalance decomposition and Brownian noise

Let $\mathcal{F}_n$ contain the history before arrival $n + 1$. Write $D_n^q = \Delta_n/q$, $h_n = n/(2q)$, and $\lambda_q = \varepsilon q$. The exact community means are

$$M_0(n)/q = h_n + D_n^q/2, \qquad M_1(n)/q = h_n - D_n^q/2.$$

For independent product-cross candidates, define

$$g_n^{\mathrm{act}} = \mathbb{P}(L_0 \le R_1 \mid \mathcal{F}_n) - \mathbb{P}(L_1 \le R_0 \mid \mathcal{F}_n).$$

Every increment $\xi_{n+1} = \Delta_{n+1} - \Delta_n$ is ± 1. The internal contributions to its mean cancel. The two cross directions give

$$b_n = \mathbb{E}(\xi_{n+1} \mid \mathcal{F}_n) = \varepsilon g_n^{\mathrm{act}}, \qquad |b_n| \le \varepsilon.$$

Define the discrete martingale $\mathcal{M}_k = q^{-1} \sum_{n<k} (\xi_{n+1} - b_n)$. Apply the martingale functional central limit theorem (FCLT) to the step process $\mathcal{M}_{\lfloor tq^2 \rfloor}$. Its maximum jump is at most $2/q$, and

$$\langle \mathcal{M} \rangle_q(t) = q^{-2} \sum_{n < \lfloor tq^2 \rfloor} (1 - b_n^2), \quad \sup_{t \le T} |\langle \mathcal{M} \rangle_q(t) - t| \le q^{-2} + T\varepsilon^2.$$

Bracket jumps are at most q^{-2}. These deterministic bounds verify the three conditions of [14, Theorem 2.1(ii)], including the expected squared-jump condition. Thus the step martingale converges to standard Brownian motion. Let M_q denote its linear interpolation. The interpolation changes it by at most $2/q$. Since the limit is continuous, M_q converges weakly in $C([0,T])$ with its uniform norm. Discrete Doob also gives

$$\mathbb{E}\|M_q\|^2_{\infty,[0,T]} \le 4(T + q^{-2}). \tag{5.6}$$

■ 5.4 Deterministic averaging of the common phase

Integer optimal couplings give

$$|g_n^{\mathrm{act}} - g(h_n, D_n^q)| \le W_n.$$

Indeed, for integer arguments an indicator $\mathbf{1}_{\{l\le r\}}$ can change by at most the sum of the integer displacements. Couple each of the four side marginals independently with its orbit target. Equations (5.5) and $\lambda_q \le \Lambda$ imply

$$\mathbb{E}\sup_{t\le T}\left|\frac{\varepsilon}{q}\sum_{n<\lfloor tq^2\rfloor}\{g_n^{\mathrm{act}} - g(h_n, D_n^q)\}\right| \le \Lambda q^{-2}\sum_{n<Tq^2}\mathbb{E}W_n \longrightarrow 0.$$

The function $g(h,d)$ is one-periodic in h, by integer translation of both orbit laws. Partition arrival indices into blocks of length exactly $2q$. For a block starting at j, discrete Doob and $|b_n| \le \varepsilon$ give

$$\mathbb{E}\max_{0\le r\le 2q}|D^q_{j+r} - D^q_j| \le 2\sqrt{2/q} + 2\varepsilon.$$

Freeze D_n^q at the block start. The phases $h_n \bmod 1$ cover the $2q$ equally spaced points of one period. Equation (4.8) bounds their Riemann-sum error by $2/q$, uniformly in the frozen imbalance. The Lipschitz constants of g and $\overline{g}$ in d are both at most two. Freezing and unfreezing therefore cost at most four times the displayed block variation. A final incomplete block contributes at most $4/q$ after normalization. Hence

$$\begin{aligned}\mathbb{E}\sup_{t\le T}\left|q^{-2}\sum_{n<\lfloor tq^2\rfloor}[g(h_n, D_n^q) - \overline{g}(D_n^q)]\right| \\ \le C(T+1)(q^{-1/2} + \varepsilon + q^{-1}) \longrightarrow 0.\end{aligned} \tag{5.7}$$

The common phase traverses one period every $2q$ arrivals.

Let D_q be the linear interpolation of D_n^q at times n/q^2. Combining the last two estimates, the exact martingale decomposition, and a Riemann sum gives

$$\begin{aligned}D_q(t) = M_q(t) + \lambda\int_0^t \overline{g}(D_q(s))\,\mathrm{d}s + R_q(t), \\ \mathbb{E}\|R_q\|_{\infty,[0,T]} \longrightarrow 0.\end{aligned} \tag{5.8}$$

The remaining errors are bounded by $T|\lambda_q - \lambda|$, $O_T(\Lambda/q)$ for interpolation inside the drift integral, and $O(1/q)$ for fractional final steps. These bounds use $|\overline{g}| \le 1$ and its global Lipschitz constant two.

For a continuous driver m, let $\Gamma_\lambda(m)$ solve

$$x(t) = m(t) + \lambda\int_0^t \overline{g}(x(s))\,\mathrm{d}s.$$

Picard iteration gives existence and uniqueness. Gronwall gives

$$\begin{aligned}\|\Gamma_\lambda(m) - \Gamma_\lambda(\widetilde{m})\|_\infty \le e^{2\lambda T}\|m - \widetilde{m}\|_\infty, \\ \|D_q - \Gamma_\lambda(M_q)\|_\infty \le e^{2\lambda T}\|R_q\|_\infty.\end{aligned}$$

The weak convergence of M_q, together with (5.6), implies first Wasserstein convergence. For example, use a Skorokhod representation in the Polish space $C([0,T])$. Uniform second moments make the represented uniform distances uniformly integrable. Their almost sure convergence to zero then gives convergence in mean. Applying the Lipschitz map Γ_λ and (5.8) proves first Wasserstein convergence of D_q to the solution of (2.4). This also identifies Brownian variance one.

The other drift claims in Theorem 2.3 follow from (4.7). The density w has mass one and is strictly positive. The front F is smooth and strictly decreasing, with bounded derivatives of every fixed order. Differentiation under the integral is therefore valid. In particular,

$$\overline{g}'(d) = 2\int_{\mathbb{R}} w(r)\{F'(r+1+2d) + F'(r+1-2d)\}\,\mathrm{d}r < 0.$$

Oddness, negativity for $d > 0$, and the far-field sign limits follow directly. Equation (4.9) gives $-\overline{g}'(0) > 0$. The marginal coupling estimate gives the global Lipschitz bound.

■ 5.5 The endpoint curve, including its joint first moment

The positive-part functional along the orbit satisfies

$$\sigma'(u) = \sum_{k\ge 0}\{a_k y_k + b_k x_{k+1}\} = x_0(u)y_0(u). \tag{5.9}$$

The sum telescopes. Equivalently, a positive-part increment occurs when both offered loads are nonnegative. The tails increase with u. Thus σ is convex, nondecreasing, and 1-Lipschitz. Equation (2.3) gives evenness and convexity in d, the stated 1-Lipschitz bound in d, and the 2-Lipschitz bound in h. Full support gives $S_0(0) > 0$.

The numerical error budget uses a sharper bound in d. Since $0 \le \sigma' \le 1$, differentiation gives

$$\partial_d S_h(d) = \tfrac{1}{2}\{\sigma'(h + d/2) - \sigma'(h - d/2)\}, \qquad |\partial_d S_h(d)| \le \tfrac{1}{2}.$$

Thus the exact readout is $1/2$-Lipschitz in d, including at $h = 0$.

Fix $H_0 < \infty$ and write $\mathcal{S}(d) = (S_h(d))_{|h|\le H_0}$ for the overflow curve. Write $\kappa_q = n_*/q^2 = 2c_q/q \to \kappa > 0$ and choose a path horizon $T_* > \kappa$. Increase T_* if a longer joint path is requested. All window times belong to $[0, T_*]$ for large q.

Define the continuous curve F_q by linear interpolation of

$$F_q\left(\frac{k - n_*}{2q}\right) = \frac{Z_{c_q}(k)}{q}.$$

It is deterministically 2-Lipschitz on $[-H_0, H_0]$. Its distance from the original floor-indexed curve is at most $1/q$. This specifies the interpolation in Theorem 2.3.

At a fixed h, put $n = n_q(h)$, $h_q = (n - n_*)/(2q)$, and $\tau_q(h) = n/q^2$. Then $|h_q - h| \le 1/(2q)$. Integer translation by c_q and positive-part Lipschitz continuity give the exact comparison

$$\left|\frac{Z_{c_q}(n)}{q} - S_{h_q}(D_q(\tau_q(h)))\right| \le W_n/2. \tag{5.10}$$

The factor $1/2$ comes from $m/q = 1/2$ for each side. Since $\tau_q(h) \to \kappa > 0$, the expectation of W_n tends to zero by (5.5).

We also need to move the slow endpoint from $\tau_q(h)$ to κ. Put

$$a_q = |\kappa_q - \kappa| + 2H_0/q + 2/q^2 \longrightarrow 0.$$

All these times lie within a_q of κ. A martingale oscillation bound on that deterministic interval, followed by the bounded drift estimate, gives

$$\mathbb{E} \sup_{|h| \le H_0} |D_q(\tau_q(h)) - D_q(\kappa)| \le C\sqrt{a_q + q^{-2}} + C\Lambda a_q + C/q \longrightarrow 0. \tag{5.11}$$

For this bound, start the martingale at the left endpoint of the interval. Doob controls its maximum oscillation by the square root of the interval length. Taking differences through that endpoint controls every requested pair of times. There is no rate assumption on $c_q/m \to \kappa$.

Let $G_q(h) = S_h(D_q(\kappa))$. Combining (5.10), (5.11), and the interpolation bounds gives

$$\mathbb{E}|F_q(h) - G_q(h)| \longrightarrow 0 \quad \text{for every fixed } h.$$

Both F_q and G_q are 2-Lipschitz. For any fixed mesh with maximum gap at most η,

$$\|F_q - G_q\|_\infty \le \max_{h \text{ in the mesh}} |F_q(h) - G_q(h)| + 4\eta.$$

Take expectations, then $q \to \infty$, then $\eta \downarrow 0$. The finite maximum is bounded by the sum of its errors. We obtain the stronger, same-process estimate

$$\mathbb{E}\|F_q - \mathcal{S}(D_q(\kappa))\|_{\infty,[-H_0,H_0]} \longrightarrow 0. \tag{5.12}$$

The finite mesh upgrades the deterministic-time estimates to uniform control of the overflow curve.

The map $f \mapsto (f, \mathcal{S}(f(\kappa)))$ is Lipschitz from the uniform path norm to the sum of the two uniform norms. First Wasserstein convergence of D_q and (5.12) therefore prove the claimed joint path-and-curve convergence. Restriction gives the result on any shorter requested path horizon. In particular, the conclusion controls both the law and expected critical overflow.

Finally, couple any admissible fixed cross kernel with the product process using Theorem 2.1 on $[0, T_* q^2]$. The joint path-and-curve transport cost is at most twice $\max_n \mathcal{D}_n/q$, up to vanishing interpolation errors. Its expectation tends to zero. This transfers the first Wasserstein conclusions and finishes Theorem 2.3.

■ 6 Evaluation of the critical law

The experiments assess the two prediction steps separately. The readout comparison conditions on observed imbalance to isolate internal variation. The terminal comparison assesses the diffusion and unconditional overflow mean. All evaluations use preserved samples and frozen lookup tables. The profile readout has no fitted parameter.

■ 6.1 Certified evaluation

The profile has no closed form here. An a posteriori certificate evaluates it from a saved probability trajectory, with binary64 inputs treated as exact rationals. It combines a full polynomial residual, analytic infinite-tail bounds, and lower transition kernels for the midpoint difference equation. The uniform profile error is $D_* < 9.038\,10^{-6}$. The resulting certified interval

$$S_0(0) \in [0.63127654,\ 0.63129462]$$

shows the size of the internal contribution even at zero imbalance. The restoring coefficient satisfies $-\overline{g}'(0) \in [0.71997987,\ 0.72012468]$. Supplement sections SM3.1 and SM4 give the construction and all reported readout bounds.

Fig. 3 shows the two-side profile and averaged response from the saved numerical nodes. The different side laws record the tie rule. Their half-phase shift agrees with the identity proved in Section 4. The response points toward zero imbalance and saturates at large imbalance, as required by Theorem 2.3.

■ 6.2 The internal contribution at observed imbalance

Use the prospective batch described in Section 6.4. Its four cells have $q = 512$, $\kappa = 1$, $\lambda = 1, 2$, and matching or product crosses, with 4096 shared replicate labels per cell. For each saved endpoint at $n_* = q^2$, compare

$$x = Z_c(n_*)/q, \qquad b = |\Delta_{n_*}|/(2q), \qquad p = S_0(\Delta_{n_*}/q).$$

Fig. 1 compares the two readouts at the same observed imbalance. Across all four cells, mean absolute error falls from 0.3961–0.4355 to 0.0174–0.0180, a 95.6–95.9% reduction. These paired errors use the same endpoints as the terminal comparison. The configuration audit and treatment of shared labels are in Supplement sections SM7 and SM5.

■ 6.3 Finite-scale errors

The original batch uses $q = 64, 128, 256, 512$, $\lambda = 0, 1, 2$, and matching or product crosses. It has respectively 128, 128, 256, and 1024 labels per cell. Feedback error compares the actual conditional cross response with $g(h_n, \Delta_n/q)$ at 16 prescribed phases. Overflow error is the maximum of $|Z_c/q - S_h(\Delta/q)|$ over the five prescribed offsets $h = -1, -1/2, 0, 1/2, 1$. Fig. 5 summarizes the nonzero-coupling cells.

At $q = 512$, the frozen inference supports population failure probability at most 0.10 for feedback margin 0.10 and overflow margin 0.05. Overflow failures remain, with probability upper bounds 0.058–0.073. The original cross-kernel cumulative distribution function (CDF) bounds are at most 0.130602. Supplement section SM5.1 gives every primary cell and the separate joint error allocation of at most 0.05.

■ 6.4 Terminal diffusion and unconditional overflow

A separately frozen prospective batch has $q = 512$, $\kappa = 1$, $\lambda = 1, 2$, and matching or product crosses. Its four cells each have 4096 labels. The labels are shared across cells, giving 16384 endpoints. Each label

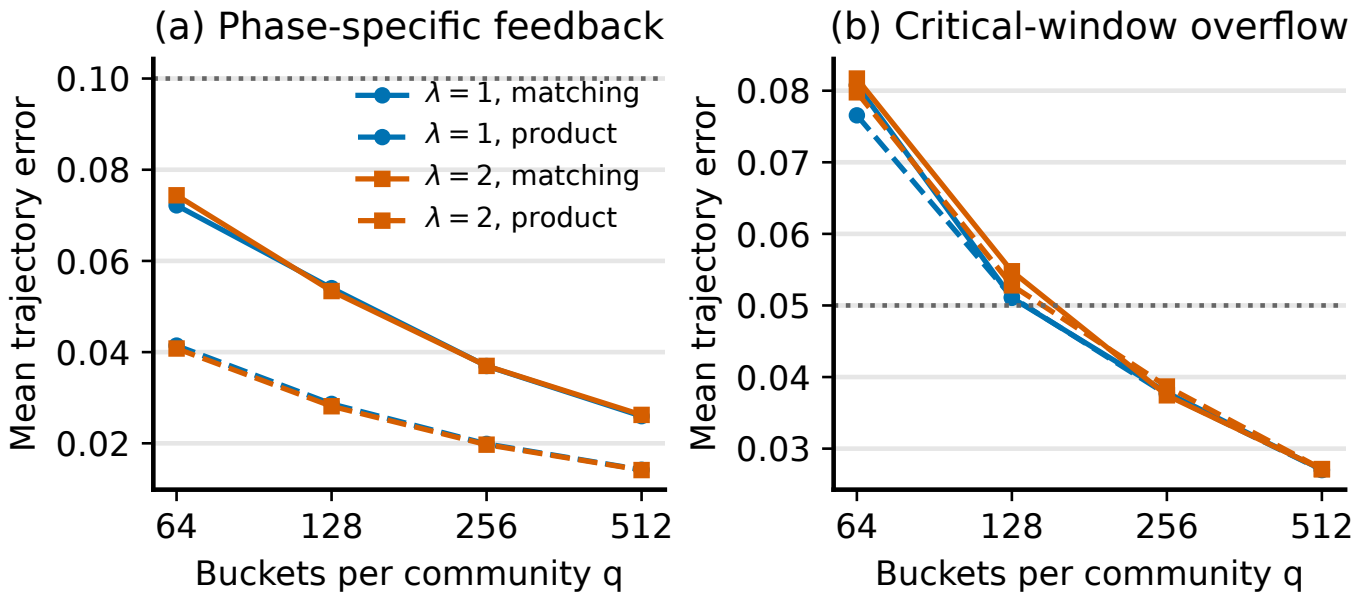


Fig. 5 Finite-scale errors from the original saved outcomes. Left: mean per-trajectory feedback root mean square (RMS) over 16 phases. Right: mean per-trajectory maximum overflow error over five offsets. Line styles distinguish matching and product kernels; markers distinguish $\lambda = 1, 2$. The horizontal lines mark the frozen per-trajectory margins. Each cell is summarized separately.

is the sampling unit. The targets are the unconditional mean difference $\mathbb{E}[Z_c(q^2)/q] - \mathbb{E}[S_0(D_1)]$ and the Kolmogorov distance between $\mathcal{L}(\Delta_{q^2}/q)$ and $\mathcal{L}(D_1)$. The reference uses 32768 preserved stochastic differential equation (SDE) paths per λ, with 16384 Euler steps per path. No observed microscopic imbalance enters its overflow mean.

All mean intervals lie within the frozen $[-0.05, 0.05]$ margin. All CDF upper bounds lie below the frozen 0.20 margin. The eight comparisons have joint failure allocation below 0.018138367441, giving at least 95% coverage with the conditions collected in Section 7.1. Coverage is unconditional over the reused reference and fresh data, in a separate family from the original comparisons.

7 Discussion

Community excess omits a strictly positive order-q internal overflow contribution. The profile readout retains it. Corollary 2.4 identifies the missing term, and Corollary 2.5 gives the resulting leading overflow cost. Every admissible fixed cross kernel shares that cost. Replacing a fixed matching by product sampling therefore preserves the leading critical overflow law.

7.1 Scope and limitations

The analysis uses complete-bipartite internal sampling, predetermined doubly stochastic cross kernels, integer loads, and left-favored ties. Extensions to other internal graphs, adaptive sampling, and other tie rules are natural further questions.

Theorem 2.1 provides a conservative bound for asymptotic comparison of the fixed-kernel and product processes. An effective uniform finite-size error bound for the joint critical limit remains open. The numerical comparisons provide supporting evidence at the tested parameters. The deterministic certificates bound evaluation errors in the limiting profile and its readouts; they do not bound finite-size allocation errors. Quantitative rates for profile attraction and tracking are natural next steps.

The reference-error and coverage guarantees use the arithmetic and ideal independent-word assumptions specified in Supplement sections SM6 and SM5. The word model assumes independence within streams and across labels, with fresh streams independent of prior history.

Acknowledgements

This work was supported by the National Natural Science Foundation of China under Grant No. U25A20426, "Key technologies enabling trustworthy, secure, and efficient multi-modal financial data factor circulation."

Competing interests

The authors declare that they have no competing interests or financial conflicts to disclose.

Data availability

The data and code supporting this study are publicly available at `https://github.com/19991111/critical-overflow-reproducibility`. Supplement section SM7 identifies the frozen inputs, original source commits, and reproducibility records.

Declarations

OpenAI's Codex assisted with literature searches, manuscript organization and drafting, proof checking, code preparation for reanalysis of preserved outputs, figure preparation, and document checks. The authors assume responsibility for all content.

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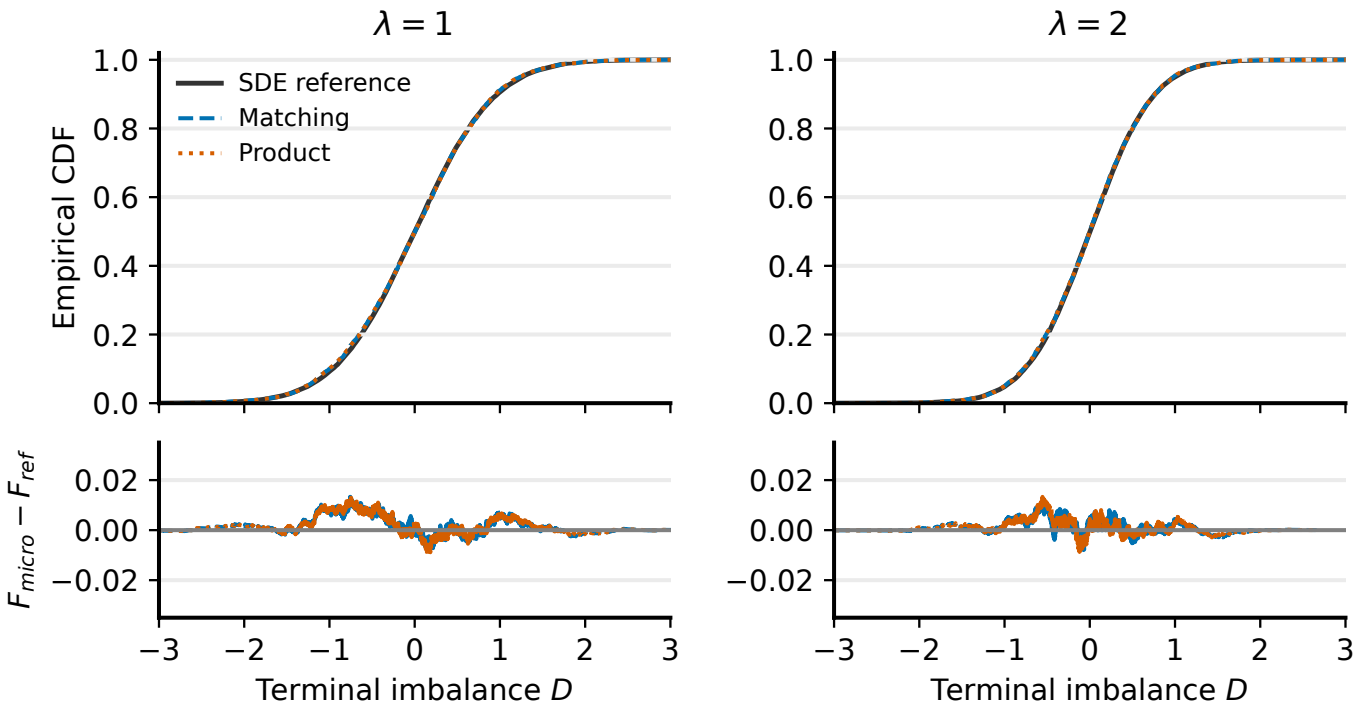


Fig. 6 Terminal imbalance: the two prospective microscopic empirical CDFs and the independently generated, preserved SDE reference for each λ. The lower panels show microscopic minus reference empirical CDFs at all jumps. Each microscopic cell has 4096 labels; each reference has 32768 paths. The numerical and sampling allowances enter the distance bounds in Table 1.

Table 1 Prospective comparisons at $q = 512$, $\kappa = 1$. Mean intervals bound microscopic mean overflow minus the SDE mean readout. CDF upper bounds concern terminal imbalance and have lower bound zero. All displayed limits are rounded outwards and include reference error.

λ	Cross kernel	Mean-difference interval	CDF upper bound
1	matching	$[-0.00840,\ 0.01010]$	0.06542
1	product	$[-0.00860,\ 0.00993]$	0.06500
2	matching	$[-0.00663,\ 0.00844]$	0.07021
2	product	$[-0.00671,\ 0.00834]$	0.07119

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